If you forget formula of any boolean function then:
Draw T.T (Truth table).
Eg: XNOR → (XOR)' → A B | A⊕B | $\bar{A}\bar{B}+AB$

Date / /
Page

Implement all Gates using NAND & NOR Gate

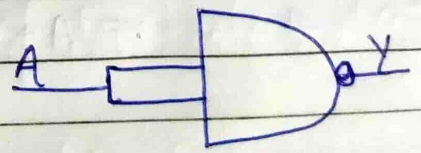
Nand & NOR gates are Universal gates.

A universal gate is one with which we can implement all basic logic gates (AND), (OR), (NOT) and therefore any digital circuit.

Gates	No. of NAND Gates required	No. of NOR gate required
NOT	1	1
AND	2	3
OR	3	2
XOR	4	5
XNOR	5	4

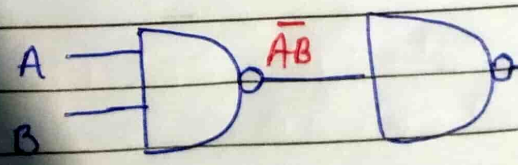
→ No. of NAND Gates required to make

(i) NOT



$[Y = \bar{A}]$

(ii) AND



$\overline{(\overline{AB})} = AB$

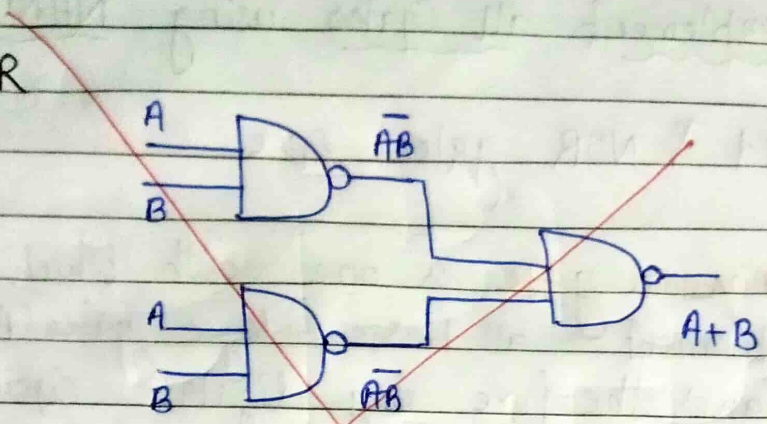
$\overline{(\overline{AB})} = AB$

$[(A')' = A]$

Akash

Date : / /
Page :

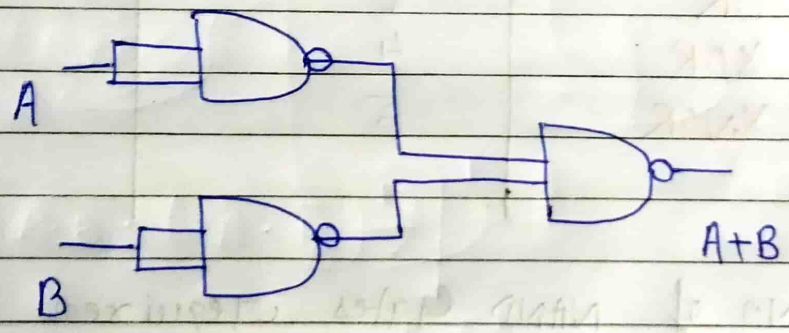
(iii) OR



$$\left[(\overline{AB})\cdot(\overline{AB})\right]'$$

A

(iii) OR



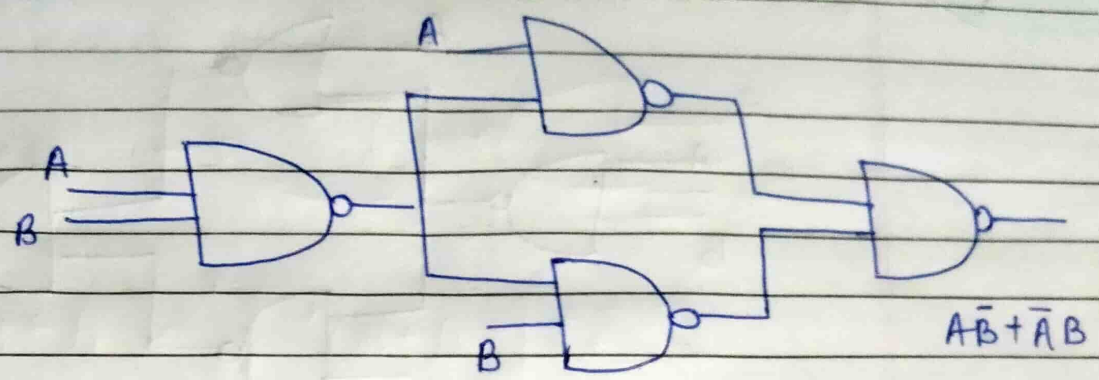
$$(\overline{A}\cdot\overline{B})' = (A')' + (B')'$$

$$= A + B \qquad \left[(A\cdot B)' = A' + B'\right]$$

Akash

Date ___/___/___
Page _____

(iv) XOR :-



$$\overline{AB}\cdot\left[(\bar{A}\cdot\bar{B})'\right]$$

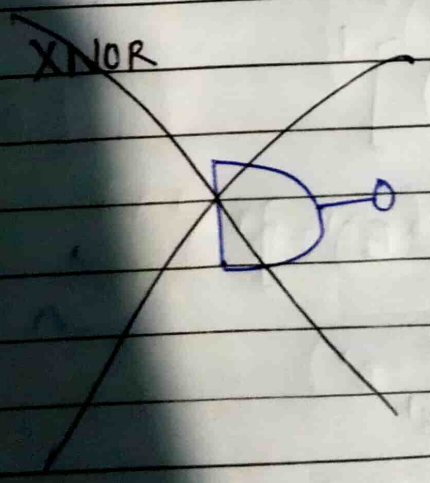
$$(\bar{A}+\bar{B})\cdot(A+B)$$

$$A\bar{A}+A\bar{B}+\bar{A}\cdot B+B\cdot\bar{B}$$

$$0+A\bar{B}+\bar{A}\cdot B+0$$

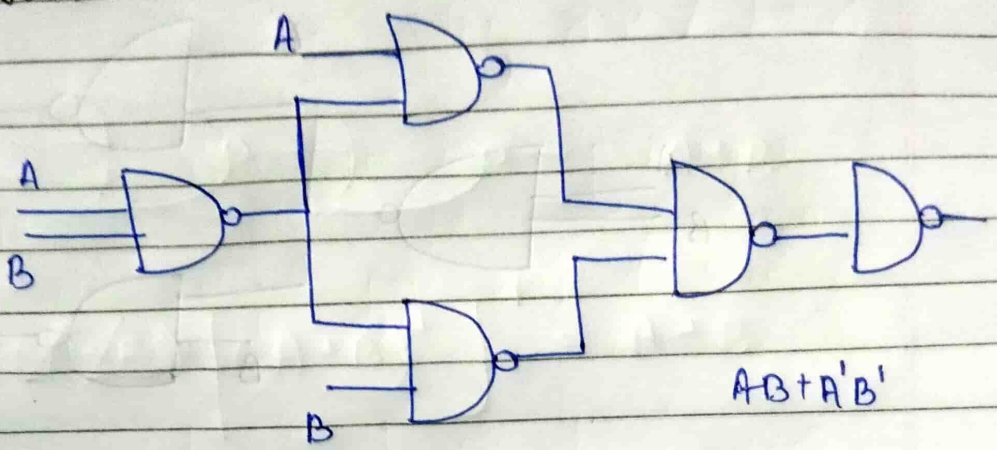
$$A\bar{B}+\bar{A}B$$

(v) XNOR



PYQ-23-24

(v) XNOR



$$\left[\overline{A\cdot B}\cdot(\overline{A\cdot B})'\right]'$$

$$\left[\overline{A\cdot B}\cdot(\bar{A}\cdot\bar{B})'\right]'$$

$$\left[(\bar{A}+\bar{B})\cdot(A+B)\right]'$$

$$\left[\overline{A\cdot B}=A'+B'\right]$$

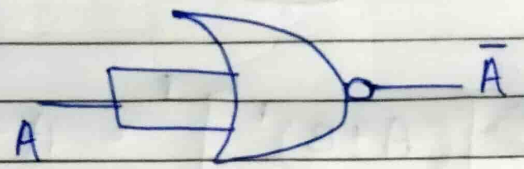
$$\left[(\bar{A}+\bar{B})'+(A+B)'\right]$$

$$AB+A'B'$$

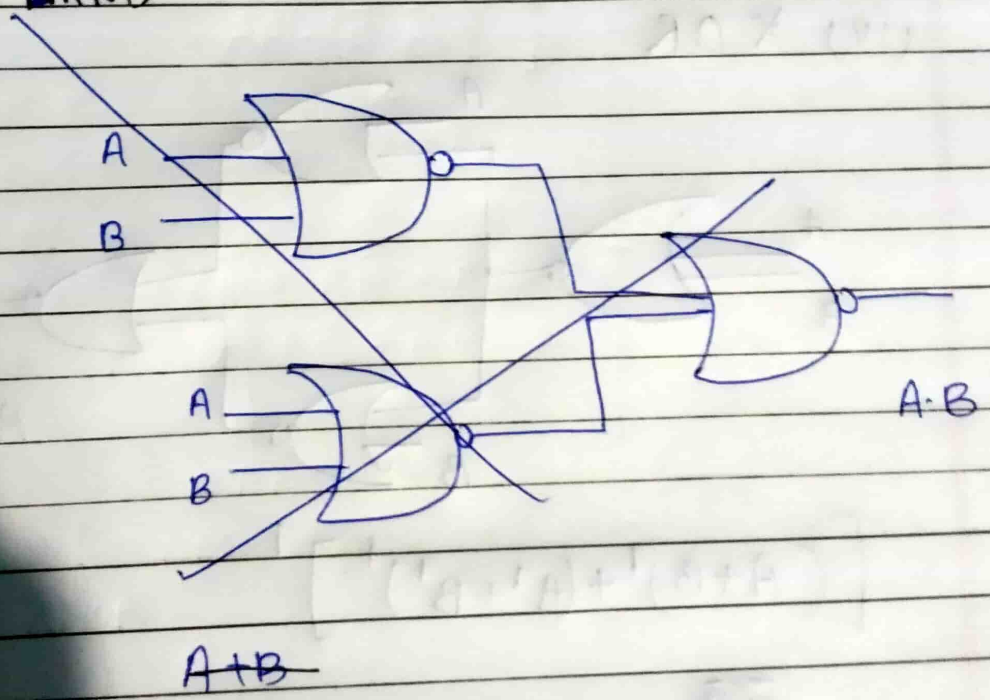
Akash

→ No. of NOR Gates required to make

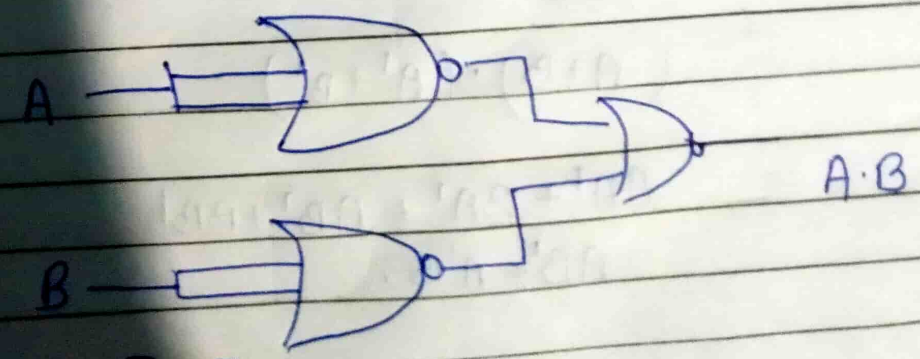
(i) NOT



(ii) AND



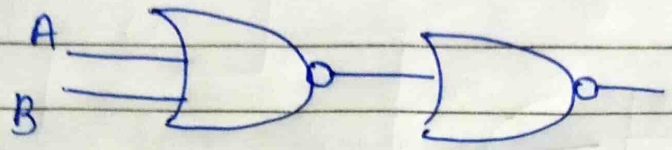
(ii) AND



$(\bar{A}+\bar{B})' = AB$

Akash

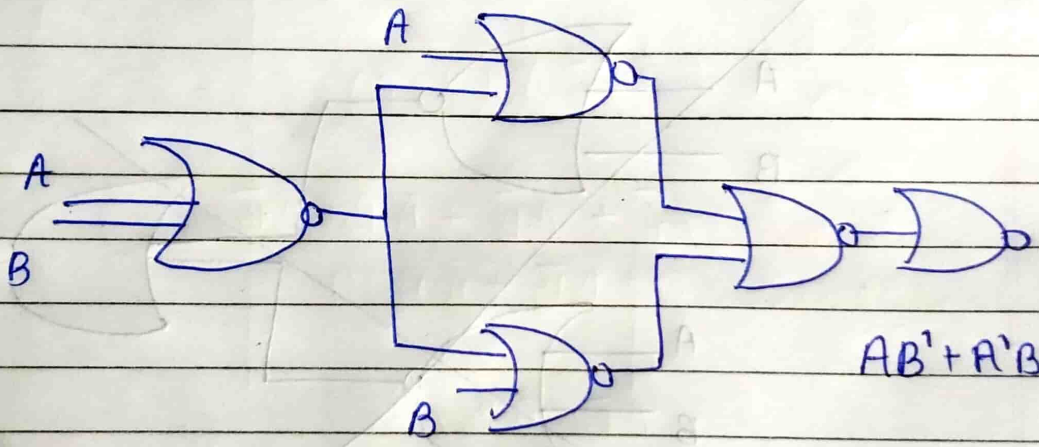
(iii) OR



$A+B$

$$\left[(A+B)'\right]' = A+B$$

(iv) X OR



$AB'+A'B$

$$\left[(A+B)'+(A'+B')'\right]'$$

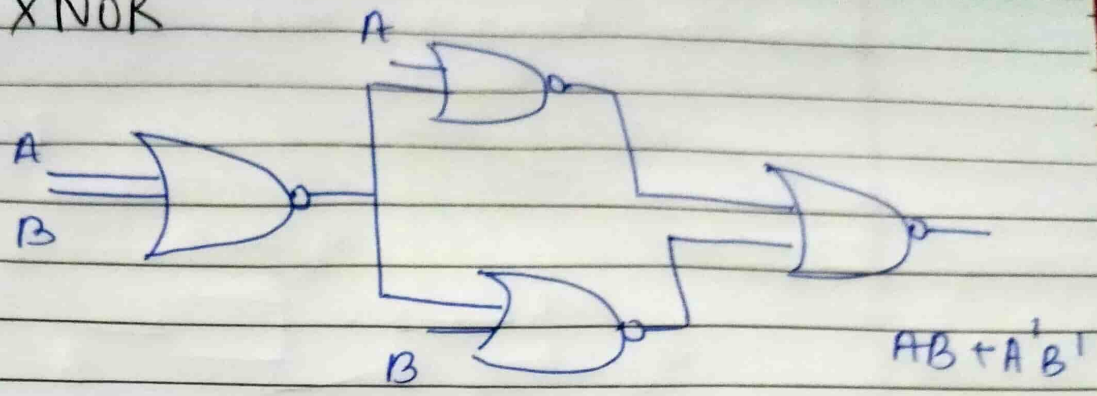
$$((A+B)')'\cdot((A'+B')')'$$

$$(A+B)\cdot(A'+B')$$

$$AA'+BA'+AB'+BB'$$

$$AB'+A'B.$$

(v) XNOR



$(A+B)' + (A'+B')'$

$A'B' + AB$ //.

Akash